TOR AMPLITUDE OF DERIVED QUASI-COHERENT OBJECTS TESTED ON QUASI-COHERENT MODULES OVER QUASI-SEPARATED SCHEMES

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Allowed standard prerequisites (editorial).

- For a quasi-compact separated morphism of schemes the direct image preserves quasi-coherence, whence the restriction functor on quasi-coherent modules to an affine open of a quasi-separated scheme is essentially surjective. invoked at proof lines 2615-2620. Transfers the vanishing condition from X to the affine open U.
- On an affine open the derived quasi-coherent object corresponds to a complex of modules over the ring. invoked at proof lines 2624-2626. Reduces the tor amplitude to the module level.
- A complex of modules whose derived tensor products with all modules have cohomology in a fixed range has tor amplitude in that range, by the affine tor dimension lemma. invoked at proof lines 2630-2632. Concludes the affine case and the lemma.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Cited at proof lines 2616-2618 for the essential surjectivity of the restriction; the transfer of the vanishing is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 2624-2626 to represent E|_U by M^bullet; the tor amplitude conclusion is inside the counted span.

Source reference. Original lemma 08E9: sources/stacks-perfect/pinned-original-source.tex, source lines 2508–2508. Cited at proof lines 2630-2632 to conclude on the affine open; the final assembly is inside the counted span.

Lemma 0.2. *Let X be a quasi-separated scheme. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Let $a \leq b$. The following are equivalent*

(1) *E has tor amplitude in $[a, b]$, and*
(2) *for all $\mathcal{F}$ in $QCoh(\mathcal{O}_X)$ we have $H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{F}) = 0$ for $i \notin [a, b]$.*

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Proof. It is clear that (1) implies (2). Assume (2). Let $U \subset X$ be an affine open. As X is quasi-separated the morphism $j : U \to X$ is quasi-compact and separated, hence j_* transforms quasi-coherent modules into quasi-coherent modules (Schemes, Lemma 01LC). Thus the functor $QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_U)$ is essentially surjective. It follows that condition (2) implies the vanishing of $H^i(E|_U \otimes^{\mathbf{L}}_{\mathcal{O}_U} \mathcal{G})$ for $i \notin [a, b]$ for all quasi-coherent $\mathcal{O}_U$-modules $\mathcal{G}$. Write $U = \mathrm{Spec}(A)$ and let $M^\bullet$ be the complex of A-modules corresponding to $E|_U$ by Lemma 06Z0. We have just shown that $M^\bullet \otimes^{\mathbf{L}}_A N$ has vanishing cohomology groups outside the range $[a, b]$, in other words $M^\bullet$ has tor amplitude in $[a, b]$. By Lemma 08E9 we conclude that $E|_U$ has tor amplitude in $[a, b]$. This proves the lemma. □